I hope this message finds you well. As we prepare to move into phase two of our project, I have conducted a comprehensive review of the three datasets:
'Transactions', ’CustomerDemographic', and ‘CustomerAddress'. Our primary objective was to assess data quality and identify any issues that may impact the integrity of our analysis. Here is a summary of our findings, assumptions, and recommendations:

1. ' Transactions'' Dataset:

Missing Values:
'online_order': 360 missing values
'brand', 'product_line', 'product_class', 'product_size', 'standard_cost', 'product_first_sold_date': 197 missing values each
Assumptions:

Missing 'online_order' values may indicate offline transactions, but we recommend confirming this assumption with you.
The missing values in product-related columns may limit our product analysis.
Recommendations:

Collaborate with you to complete missing data, especially in product-related columns.
Verify the accuracy of the assumption about missing 'online_order' values.

2. ‘CustomerDemographic’ Dataset:

Missing Values:
'last_name': 125 missing values
'DOB': 87 missing values
'job_title': 506 missing values
'job_industry_category': 656 missing values
'default': 302 missing values
'tenure': 87 missing values
Assumptions:

The 'default' column seems to contain garbled data and may not be useful for analysis.
We assume that 'gender' should ideally have values 'Male' and 'Female,' but there are inconsistent entries.
Recommendations:

Given the lack of meaningful data in the 'default' column, consider removing it.
Address missing values in 'last_name,' 'DOB,' 'job_title,' 'job_industry_category,' and 'tenure.'
Standardize 'gender' values for consistency.
3. ‘CustomerAddress’ Dataset:

No missing values detected.

The 'country' column contains only one value, 'Australia.'
'state' column values have variations like 'NSW,' 'VIC,' 'New South Wales,' and 'Victoria.'
Assumptions:

The 'country' column is redundant as it contains a single value, 'Australia.'
We assume that 'NSW' corresponds to 'New South Wales' and 'VIC' corresponds to 'Victoria.'
Recommendations:

Remove the 'country' column as it adds no value.
Standardize 'state' values to 'NSW' and 'VIC.'
These observations and recommendations are vital to ensure the quality of our data analysis in phase two.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
transaction_id product_id customer_id transaction_date online_order order_status  brand product_line product_class product_size list_price standard_cost product_first_sold_date
0              1       2950        2150       2017-02-25           1        Approved  Solex   Standard    Medium     Medium    71.19     37.97                 41064.0
1              2       3026        2067       2017-05-21           0        Approved  Giant Bicycles  Standard    Medium     large  2091.47    388.92                 41790.0
2              3       2502        3372       2017-10-16           0        Approved  WeaTherTech  Standard    Medium     Medium  1035.90    670.36                 41820.0
3              4       2035        2618       2017-08-31           1        Approved  Trek Bicycles  Standard    medium     large  1465.09    731.54                 41659.0
4              5       4811        2084       2017-10-01           0        Approved  Solex   Standard    medium     Medium   871.99    260.09                 41671.0
```

```
transaction_id  product_id  customer_id  transaction_date  online_order  order_status
count    20000.000000  20000.000000  20000.000000          19803.000000  20000.000000
mean     10000.500000   2738.064500   2847.686200           0.504217      0.993700
std       5773.819501   1843.826988   2044.251353           0.499977      0.079346
min          1.000000      0.000000      1.000000           0.000000      0.000000
25%       5000.750000   1060.000000   954.000000           0.000000      1.000000
50%      10000.500000   2505.000000   2804.000000           1.000000      1.000000
75%      15000.250000   4039.000000   4708.000000           1.000000      1.000000
max      20000.000000   5050.000000   5034.000000           1.000000      2.000000
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20000 entries, 0 to 19999
Data columns (total 26):
 #   Column                               Non-Null Count  Dtype
---  ------                               --------------  -----
 0   transaction_id                       20000           int64
 1   product_id                           20000           int64
 2   customer_id                          20000           int64
 3   transaction_date                     20000           datetime64[ns]
 4   online_order                         19803           float64
 5   order_status                         20000           object
 6   brand                                19772           object
 7   product_line                         19772           object
 8   product_class                        19772           object
 9   product_size                         19772           object
 10  list_price                           20000           float64
 11  standard_cost                        19772           float64
 12  product_first_sold_date              19772           float64
dtypes: int64 (5), float64 (5), object (12), datetime64[ns] (1)
memory usage: 2.0 MB
None
```

```
transaction_id                       0
product_id                           0
customer_id                          0
transaction_date                     0
online_order                       197
order_status                         0
brand                              228
product_line                       228
product_class                      228
product_size                       228
list_price                           0
standard_cost                      228
product_first_sold_date            228
dtype: int64
```

```
transaction_id                       0
product_id                           0
customer_id                          0
transaction_date                     0
online_order                         0
order_status                         0
brand                              197
product_line                       197
product_class                      197
product_size                       197
list_price                           0
standard_cost                      197
product_first_sold_date            197
dtype: int64
```

```
transaction_id                       0
product_id                           0
customer_id                          0
transaction_date                     0
online_order                         0
order_status                         0
brand                                0
product_line                         0
product_class                        0
product_size                         0
list_price                           0
standard_cost                        0
product_first_sold_date              0
```